INVITED PAPER

Design of Flower Constellations for Telecommunication Services

Design and optimization of systems for maximizing global coverage and network connectivity via intersatellite links and improving their performance are discussed in this paper.

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ABSTRACT | Satellite constellation designers take into consideration the entire telecommunication network and their design choices are influenced by many factors including: number of satellites, orbital characteristics, coverage area, network interconnections, system cost, and complexity. When the satellite constellation design method does not include a large number of possible configurations, then the final result of the design process is a suboptimal solution. Flower Constellation (FC) design provides a relatively new design approach which can overcome this problem. The time evolution of the FC theory is here summarized and the fundamental mathematics allowing the constellation design is provided. In particular, the theory is applied to the design and optimization of constellations maximizing the global coverage and the network connectivity via intersatellite links. Performance results are compared to the classical type of satellite constellations, i.e., the Walker Constellations. The performance improvement provided by FC design with respect to Walker Constellation design is shown. Finally, considerations regarding the cost for deployment and orbital control are also provided.

KEYWORDS | Flower Constellations (FCs); orbit control; satellite communication systems; Walker Constellations

I. INTRODUCTION

In communication services, satellite constellations are needed instead of a single satellite because a single satellite can cover only a limited portion of the Earth for some particular time intervals. On the other hand, satellite constellations can provide: simultaneous multiple coverage, continuous global coverage, continuous regional coverage, or low revisit interval. Constellation design is generally a very difficult problem because each orbit has an infinite number of choices for the six orbital parameters [1]: semimajor axis a, eccentricity e, inclination i, argument of perigee ω, right ascension of the ascending node (RAAN) Ω, and mean anomaly M. Therefore, for many satellites, the problem is of exceedingly high dimensionality. This is one of the primary reasons why the art of constellation design is presently suffering from a deep technology development delay. In order to solve this complex problem, satellite constellation designers are forced to make specific assumptions to reduce the dimensionality of the problem. These assumptions are usually limiting and preventing the discovery and development of new and useful solutions. The art of making these assumptions coincides with the time evolution of a limited number of proposed constellation design methodologies. For instance, the assumption of circular orbits, while simplifying the problem from one side, strongly limits the varieties of potential configurations on the other side. As a consequence, new design methodologies should be developed allowing to increase the set of possible solutions thus providing enhanced performance.

In general, global coverage optimality coincides with symmetric satellites distribution. Walker Constellations [2], [3] provide an excellent general design methodology for circular orbits that has been implemented in several real missions (e.g., GPS, Galileo, etc.). Flower Constellation (FC) is also a general design methodology that uses elliptical orbits [4]. The main conceptual difference between Walker and Flower is the selected space where to search the symmetry. While Walker Constellation design methodology uses the inertial space through an

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Earth-centered inertial (ECI) reference frame, FC theory looks for symmetry in rotating reference frames, such as the Earth-centered–Earth-fixed (ECEF) reference frame. The theory of FCs has changed since it was first presented as new insights and reformulations into minimal parameterizations have been discovered over time. The first section of this paper provides the time evolution of the FC theory from its first appearance [4]–[6] up to the most recent developments [7]–[9]. The theory is then used for the design and optimization of a satellite constellation for global communication services, defining requirements and objective functions and comparing the performance provided by the two optimized constellation types (Walker and Flower).

During the constellation design and optimization process, the system cost must be taken into account not only minimizing the number of satellites of the constellation, but also including the costs for system deployment and orbit maintenance. Therefore, we close the paper with a discussion on deployment and orbit control of these constellations and provide remarks and practical considerations.

The paper is organized as follows. In Section II, the theory of FCs is presented with a particular focus on its evolution and the recent updates. Section III analyzes the optimization process of FCs for communication services and compares the performance of FCs and Walker Constellations in terms of continuous global coverage and intersatellite link availability. Section IV discusses the problems related to system engineering, i.e., deployment and orbit maintenance with a focus on FCs. Finally, in Section V, conclusions are drawn.

II. EVOLUTION OF THE FLOWER CONSTELLATIONS THEORY

The theory of FCs was first presented in May 2003 at "John L. Junkins" Astrodynamics Symposium and then published a year later in [4]. The theory consists of a general methodology to design constellations of N_s satellites all belonging to the same closed-loop trajectory, called *relative trajectory*, with respect to a rotating reference frame.[1] The original theory, which is summarized in Section II-A, is then evolved over the time and a rich literature was produced describing subsequent insights and reformulations. Several progresses were made during the years and most of the needed mathematical tools are today available to design FCs. The following Sections II-A–C describe the main mathematical progresses made in the theory while time evolution and rationale are summarized in Fig. 1.

[1]Usually, the ECEF reference frame is selected as the rotating reference frame.

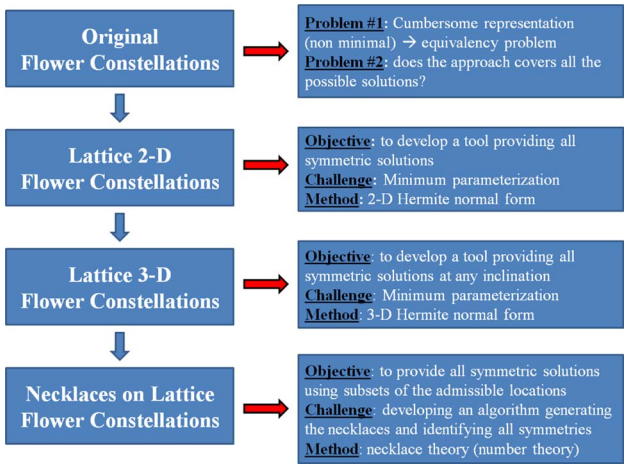

Fig. 1. ***Background rationale of Flower Constellations.***

A. Original Theory of Flower Constellations

In order to place all satellites in the same *relative trajectory*, compatible (resonant) orbits must be adopted. These orbits satisfy

$$N_p T_p = N_d T_d \tag{1}$$

stating that the orbital period T_p is a rational multiple of the period of rotation of the rotating frame (e.g., ECEF) T_d, where N_p and N_d are two positive coprime integers. Equation (1) guarantees that the trajectory in the rotating frame is closed. Two additional conditions are needed to make all the satellites following the same relative trajectory. These are:

1) the semimajor axis (a), the eccentricity (e), the orbit inclination (i), and perigee argument (ω) must be the same for all the orbits;
2) the mean anomaly (M_i) and the right ascension of the ascending node (Ω_i) of the ith satellite must satisfy

$$N_p \Omega_i \equiv -N_d M_i \bmod (2\pi). \tag{2}$$

These are necessary and sufficient conditions to have all the satellites on the same trajectory (see [10] for a complete proof). Equation (2) dictates the phasing of the satellites belonging to an FC. This equation allows to evaluate the set of "Ω_i" for a specific M_i and *vice versa*, the set of "M_i" for a specific Ω_i. Since the Ω_i identify the orbital planes, to obtain symmetric distribution of the orbital planes, Mortari *et al.* [4] proposed the following sequence:

$$\Omega_{i+1} = \Omega_i + 2\pi \frac{F_n}{F_d}, \qquad i = 1, \ldots, N_s \tag{3}$$

where F_n and F_d are two coprime positive integers. Using (3) in (2), we obtain the sequence of mean anomalies

$$M_{i+1} = M_i - 2\pi \frac{N_p F_n + F_d F_h(i)}{F_d N_d}, \qquad i = 1, \ldots, N_s \quad (4)$$

where $F_h(i)$ can be any sequence of integers modulo N_d (e.g., Fibonacci sequence) or just random integers in the set $\{1, \ldots, N_d\}$ or simply a constant integer. Equations (3) and (4) allow the evaluation of the Ω_i and M_i angles (satellite phasing) by recursive sequence starting from an assigned initial value (e.g., $\Omega_1 = M_1 = 0$). It is possible to prove [10] that this procedure always produces pairs (Ω_i, M_i) consistent with (2).

The original theory, as described above, has two fundamental drawbacks. The first issue is related to the fact that this constellation design theory *is not* a minimum parametrization theory. This was discovered as different combinations of the design integer parameters (phasing) were providing the same identical configuration. Consequently, what are the mathematical conditions under which two configurations were identical (equivalency problem) was a problem to be solved. Solving this problem led to the introduction of the *configuration number*

$$N_c = E_n \frac{N_p F_n + F_d F_h}{G} \mod F_d$$

where E_n and E_d are integers satisfying the Diophantine equation $E_n F_n + E_d F_d = 1$. The integer parameter N_c, which can be derived from the phasing integer parameters (F_d, F_n, F_h), univocally identifies the satellite phasing configuration. In particular, Avendaño and Mortari [10] have shown that, for $F_h = \text{const}$, the total number of satellites N_s admits an upper bound $N_s \leq N_{s\max} = N_d F_d / \gcd(N_d, N_p F_n + F_d F_h)$, and a constellation with $N_{s\max}$ satellites was named *Secondary Path* [6] and, later on, as *Harmonic FCs* (HFCs) [10]. Finally, it has been proven [10] that the satellite phasing of HFC is completely identified in the (Ω, M) space by three invariants: the number of inertial orbits F_d, the number of satellites per orbit $N_{so} = N_d / \gcd(N_d, N_p F_n + F_d F_h)$, and the configuration number $N_c \in [0, F_d)$.

The second issue with the original theory was the fact that it was not clear whether the theory encompasses all the possible configurations. These two problems were solved with the introduction of the 2-D lattice theory of FCs.

B. 2-D Lattice Theory of Flower Constellations

The 2-D lattice theory was presented in [7], by rewriting the $\Omega - M$ phasing, (3) and (4), as $\Omega_{ij} = 2\pi i / N_o$ and $M_{ij} = (2\pi j / N_{so}) - (N_c \Omega_{ij} / N_{so})$, where $N_o = F_d$, $i = 0, \ldots, N_o - 1$, and $j = 0, \ldots, N_{so} - 1$. These two equations can be written in the more compact matrix notation as

$$\begin{bmatrix} N_o & 0 \\ N_c & N_{so} \end{bmatrix} \begin{Bmatrix} \Omega_{ij} \\ M_{ij} \end{Bmatrix} = 2\pi \begin{Bmatrix} i \\ j \end{Bmatrix} \quad (5)$$

where $N_c \in [0, N_o - 1]$. Using this notation, the (i, j) satellite is the jth satellite on the ith orbital plane. In the original theory, the integers appearing in (5) were not independent as they satisfy the (unnecessary) constraint $\gcd(N_o, N_{so}, N_c) = 1$. Equation (5), rewritten in the general form

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{Bmatrix} \Omega_{ij} \\ M_{ij} \end{Bmatrix} = \mathbf{L} \begin{Bmatrix} \Omega_{ij} \\ M_{ij} \end{Bmatrix} = 2\pi \begin{Bmatrix} i \\ j \end{Bmatrix} \quad (6)$$

describes the general regular lattice of points in the $\Omega - M$ space. Since this space is modulo 2π on each axis, it geometrically represents a torus with lattice of points. This is the current mathematical representation of FCs where the constraint $\gcd(N_o, N_{so}, N_c) = 1$ has been removed. In particular, from the lattice matrix $\mathbf{L}$, we can derive the total number of satellites $N_s = |\det(\mathbf{L})|$, the number of satellites per orbit $N_{so} = |\gcd(b, d)|$, and the number of satellites with same mean anomaly $N_{sm} = |\gcd(a, c)|$. In addition, we can evaluate the number of orbits $N_o = N_s / N_{so}$ and the number of different mean anomalies $N_m = N_s / N_{sm}$. Interesting is the fact that lattice representation can be described by the minimal parameter representation which is the Hermite normal form, a 2×2 triangular matrix of integers $\mathbf{H}$, that can be derived from the fully populated lattice matrix

$$\mathbf{H} = \begin{bmatrix} N_o & 0 \\ N_c & N_{so} \end{bmatrix} = \mathbf{L}U, \qquad \text{where} \quad U \in GL_2(\mathbb{Z}). \quad (7)$$

In the 2-D lattice theory of FCs, the satellite phasing is completely independent from the orbit size and the compatibility condition, and (1) was abandoned. This implies that nonrepeating space tracks can be used without affecting the symmetries of the satellite distribution. The remaining parameters required to design a 2-D lattice FC are the same for all orbits in the constellation: the inclination angle, the eccentricity, and the argument of perigee.

In the 2-D lattice theory all the full symmetric configurations can be obtained (by changing the value of the configuration number N_c) and the theory itself includes, as a subset, the original theory. However, there is still pending over this design methodology the curse affecting all the constellation design methodologies: the Earth oblateness

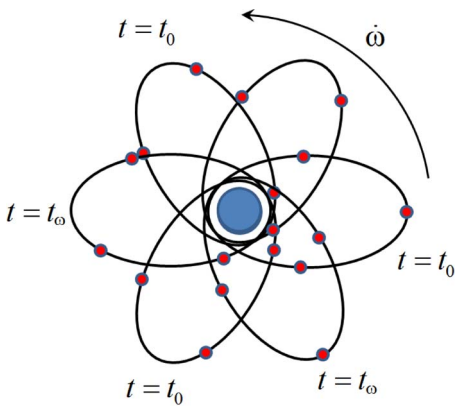

Fig. 2. ***Example of orbits rotating on the same orbital plane.***

(J_2 effect) rotates the apsidal line and, therefore, the critical inclinations must be adopted if elliptical orbits need to be used. This is the reason why today we have either constellations made of circular orbits or constellations made of elliptical orbits at critical inclination (63.4° or 116.16°). To overcome this issue the 3-D lattice theory [8] was devised.

C. 3-D Lattice Theory of Flower Constellations

The oblateness of the Earth (J_2 effect) has two main persistent effects on the orbits. In general, it causes small persistent variations of right ascension of ascending node and argument of perigee

$$\dot{\Omega} = -\frac{3}{2} J_2 \left(\frac{R_E}{p}\right)^2 n \cos i$$

and

$$\dot{\omega} = \frac{3}{4} J_2 \left(\frac{R_E}{p}\right)^2 n (5\cos^2 i - 1) \tag{8}$$

where R_E is the Earth equatorial radius, p is the semiparameter, J_2 is the spherical harmonic coefficient describing Earth oblateness, n is the modified mean motion, and i is the inclination. The $\dot{\Omega}$ effect does not represent a big issue as it can be easily compensated by small changes of the semimajor axis. In the contrary, the argument of perigee rotation (issue affecting elliptical orbits only) is a problem more difficult to solve.

The idea leading to the development of the 3-D lattice theory of FCs is the following: the argument of perigee rotation, affecting evenly distributed orbits with identical shapes (same a and e) on the same orbital plane is the same for all the orbits. Fig. 2 shows this idea for a group of $N_\omega = 3$ orbits evenly distributed on the same orbital plane at times $t = t_0$ and $t_\omega = (2\pi/\dot{\omega})(1/2N_\omega)$. This approach is particularly well suited for any mission requiring global coverage.

From a mathematical point of view it is possible to demonstrate [8], [11] that this is described by a 3-D lattice theory or, equivalently, by its minimal representation, a 3×3 Hermite normal form. The equation generating the satellite phasing will be

$$\begin{bmatrix} N_o & 0 & 0 \\ N_c^3 & N_\omega & 0 \\ N_c^1 & N_c^2 & N'_{so} \end{bmatrix} \begin{Bmatrix} \Omega \\ \omega \\ M \end{Bmatrix} = 2\pi \begin{Bmatrix} i \\ k \\ j \end{Bmatrix} \tag{9}$$

where N_o indicates the number of orbital planes, N_ω is the number of orbits per orbital plane, and N_{so} is the number of satellites per orbit. The indices in (9) belong to the following ranges:

$$i \in [0, N_{op} - 1], \; j \in [0, N_{so} - 1], \text{ and } k \in [0, N_\omega - 1]$$

while the dimensionality for the phasing gives three configuration numbers whose ranges are

$$N_{c1} \in [0, N_{op} - 1], \; N_{c2} \in [0, N_\omega - 1], \text{ and } N_{c3} \in [0, N_{op} - 1].$$

In the 3-D lattice theory, the satellites within a given orbital plane are placed in N_ω orbits with arguments of perigee evenly distributed. All the orbits have identical i, e, and a, to equal the perigee rotation rate. Most important is the fact that Davis *et al.* [8] and Davis [11] have proven that the 3-D lattice theory framework includes as subsets: 1) the original theory [4], [10], 2) the 2-D lattice theory [7], 3) the Walker Constellations [2], [3], 4) the Elliptical Walker Constellations [12], [13], and 5) the constellations developed by Draim for global coverage [14]–[16]

$$\underbrace{\begin{bmatrix} N_o & 0 & 0 \\ 0 & 1 & 0 \\ N_c & 0 & N_{so} \end{bmatrix}}_{\text{2-D Lattice Flower Constellations}} \qquad \underbrace{\begin{bmatrix} P & 0 & 0 \\ 0 & 1 & 0 \\ -F \bmod P & 0 & S \end{bmatrix}}_{\text{Walker Constellations}}$$

where the design parameters of Walker Constellations are defined as follows: P is the number of orbit planes, S is the number of satellites in each orbital plane, $T = S \cdot P$ is the number of satellites, and F is the relative spacing between satellites in adjacent planes so that if there is a satellite with mean anomaly M in one orbital plane, then in the next orbital plane there is a satellite with mean anomaly $M + 2\pi F/T$.

The lattice theories are particularly efficient for global coverage due to the uniformity of satellite distribution in the $\Omega - M$ space (2-D lattice) or $\Omega - \omega - M$ space (3-D lattice). The limitation of this theory is that for a given lattice matrix we obtain a given set of $N_s = |\det(\mathbf{H})|$ admissible positions, solutions of (5) for 2-D lattice or (9) for 3-D lattice. All these admissible locations must be filled by satellites to obtain symmetric (uniform) satellites distributions. This is a limitation because it is possible to select subset of satellites such that the symmetry is maintained, and the current solution to this problem uses the theory of necklaces [9], a special area of number theory.

III. APPLICATIONS TO TELECOMMUNICATIONS SERVICES

In this section, the design and analysis of a subset of Flower and Walker Constellations is provided with the aim to compare their performance for telecommunication services given a set of system requirements.

A. Definition of Requirements

The design of a communication system starts with the definition of the user requirements for a given type of communication service (e.g., mobile telephony services or fixed broadband communication services). User requirements include: data rate, latency, and service availability.

On the other hand, after the decision of the particular services that will be provided by the system, the designer should take into account the cost and complexity of the system in order to define the architecture requirements in terms of: type of coverage, network interconnection via intersatellite links (ISLs), number and location of network control ground stations, type of user ground terminals, type of connectivity through bidirectional/unidirectional links, number of satellites, minimum elevation angle, and communication features (antenna gain, transmit power, modulation, and coding scheme).

The Earth coverage of satellite communications services can be classified with respect to the following factors:

- *geographic scale*: the geographic scale of the coverage can be categorized into global scale (i.e., the entire Earth), regional scale (i.e., part of one or more continents), or local scale (i.e., one or more countries);
- *temporal scale*: the coverage of a given site can be provided on a continuous-time basis or with an accepted level of discontinuity;
- *multiplicity of satellites*: while the visibility of one satellite is the minimum requirement, the simultaneous visibility of a certain number of satellites can be useful for telecommunications services if satellite diversity is used to improve the performance when the conditions of the communication channel are not good or if a mobile user is located in urban areas where the probability that a satellite is shadowed is high.

The use of ISLs allows to decrease the end-to-end latency and the required number of network control ground stations for the distribution of the data traffic [17]. A large number of available ISL minimize the number of hops required to route the traffic from the source to the destination thus decreasing the end-to-end latency.

In this paper, we focus on the design of a constellation of N_s satellites for mobile and fixed communication services mainly taking into account coverage requirements and ISLs requirements for the study of an optimized MEO/LEO satellite constellation. In particular, the optimal satellite constellation should maximize at the same time the continuous global coverage and the availability of ISLs. In Section III-B, we define the parameters that will be used to build the objective function of the optimization process.

B. Design Choices and Objective Function

In order to evaluate the radio frequency visibility between a satellite and a ground terminal (the observer) with the aim to determine the coverage characteristics of the system, the Topocentric Horizon Coordinate System is used [1]. In fact, this coordinate system allows to compute the elevation angle, the azimuth, and the range of a ground terminal which points the satellite through the position vector $\mathbf{r}$ of the satellite. The origin O of this coordinate system is the ground terminal on the Earth surface. The horizontal local plane is the plane tangent to the ellipsoid of the Earth at point O. It contains the $\mathbf{s}$-axis which points south and the $\mathbf{e}$-axis which points east; the $\mathbf{z}$-axis is perpendicular to the horizontal plane and points upward towards the Zenith. Therefore, this reference frame, also called south east Zenith (SEZ) frame, is a noninertial coordinate system which rotates with the Earth. Using the SEZ frame we can define elevation, azimuth, and range by using spherical coordinates instead of Cartesian coordinates. If the elevation of the vector $\mathbf{r}$ pointing the satellite in the SEZ frame centered in the ground terminal position is greater than a minimum elevation angle $\phi_{\min}$, then the radio frequency visibility variable $\nu_s \in \{0,1\}$ is set to 1; otherwise it is set to 0; we require that at least one satellite is visible in order to have $\nu_s = 1$ (single satellite visibility). When $\phi_{\min} = 0$, the radio frequency visibility is equal to the geometric visibility; in this work, we fixed $\phi_{\min} = \pi/6$.

As previously mentioned, we consider two metrics of interest for the design of optimized satellite constellations: the percentage η_{gc} (in time and space) of the global coverage of the Earth and the percentage η_{isl} (in time) of the availability of ISLs.

In order to compute the percentage in time and space of the global coverage of the Earth, we evenly distributed N_t ground terminals on the Earth surface and we computed

the radio frequency visibility during L sample instants. As a consequence, η_{gc} can be mathematically defined as

$$\eta_{gc} = \frac{100}{N_t L} \sum_{i=1}^{N_t} \sum_{j=1}^{L} v_s^{i,j} \tag{10}$$

where the radio frequency visibility $v_s^{i,j}$ is evaluated for the ith ground terminal at the jth sample instant. In general, η_{gc} increases with the number N_s of satellites of the constellation and the altitude h of the orbits. This is not true if the satellite constellation is not properly designed.

Regarding the percentage η_{isl} in time of the availability of ISLs, we assume that two satellites can establish an ISL if geometric visibility is met between the two satellites, i.e., the link does not cross the Earth. Furthermore, we require that the link does not cross the Earth atmosphere below the altitude $h_{min} = 500$ km and we assume that the ISL antennas can point in any direction. For a constellation of N_s satellites, the number of different ISL is $N_{isl} = (1/2)N_s(N_s - 1)$, therefore η_{isl} can be mathematically defined as

$$\eta_{isl} = \frac{100}{N_{isl} L} \sum_{i=1}^{N_s-1} \sum_{j=i+1}^{N_s} \sum_{k=1}^{L} v_{isl}^{i,j,k} \tag{11}$$

where the ISL availability $v_{isl}^{i,j,k} \in \{0, 1\}$ is evaluated between the ith and the jth satellite at the kth sample instant.

We now proceed to the design of the satellite constellation that maximizes the overall performance metric η defined as the linear combination of the two single metrics: $\eta = \alpha\eta_{gc} + \beta\eta_{isl}$, where $\alpha = \beta = 0.5$. This is an example of multiobjective optimization problem.

It is worth noting that η_{gc} and η_{isl} are conflicting metrics since, in order to maximize η_{gc}, the positions of the satellites should be spaced as uniformly as possible around the Earth, while in order to maximize η_{isl} the positions of the satellites should be as close as possible.

C. Optimization, Analysis, and Comparison of Results

Now, we compare the performance of FCs and Walker Constellations. We will design FCs by using the original theory since it does not include Walker Constellations as subsets.

Walker Constellations are the classical type of satellite constellations [2], [3]. As previously introduced, a Walker Constellation is characterized by three integer parameters T, P, and F, and further three real parameters h, i, Ω_{spread}: T is the number of satellites, P is the number of orbit planes, F is the relative spacing between satellites in adjacent planes, h is the orbit height, i is the inclination, and Ω_{spread} is the constraint on the maximum spreading of the RAAN for the set of satellites. The Walker Constellation is not designed to show repeating ground track. This means that the satellites belonging to a Walker Constellation cover all the longitudes with the passing of time.

For a more fair comparison between Flower and Walker Constellations, we restrict our attention to circular orbit FCs (i.e., eccentricity $e = 0$). However, in order to design a circular orbit FC, we are no longer free to choose the perigee height h_p, while the height of the orbit h, which is now equal to both the perigee height h_p and the apogee height h_a, will depend on the design parameters N_p and N_d as follows:

$$h = \sqrt[3]{\frac{\mu N_d^2}{\omega_E N_p^2}} - R_{mean} \tag{12}$$

where μ is the gravitational coefficient, ω_E is the rotation rate of the Earth, and R_{mean} is the mean radius of the Earth.

The optimization process is based on the use of genetic algorithms. The genetic algorithm is chosen because the gradient of the objective function cannot be defined and the trend of the objective function is very complex. The evolution towards the optimal solution starts from a population of randomly generated individuals.

The genome of the genetic algorithm is the vector containing the design parameters for Flower or Walker Constellations and the objective function that is maximized is η. The genome of the genetic algorithm did not include some fixed constellation parameters: the number of satellites, the orbit height, and Ω_{spread} which was set to 2π rad. The number of satellites for each constellation is chosen on the basis of the orbit height as follows: if N_s^{min} is the minimum number of satellites required to guarantee $\eta_{gc} > 95\%$ at a given orbit height h, then the number of satellites used for the optimization was chosen as $N_s = N_s^{min} - 1$. The reason for such an approach is that it allows to optimize the satellite constellation since we are not at the performance saturation point and it allows to compare the performance of Walker Constellations and FCs when the minimization of the number of satellites is an important cost factor and can be considered as a constraint.

The detailed results of the optimization process for different orbit heights are shown in Table 1 and summarized in Fig. 3.

The comparison between Walker Constellations and FCs in terms of the maximum achievable global metric η and with the same number of satellites and the same orbit height shows that FCs provide better performance.

The higher performance is a consequence of the larger number of design parameters of FCs with respect to the number of design parameters of Walker Constellations

Table 1 Results of the Multiobjective Optimization Problem. Performance of Walker and Flower Constellations Can be Compared Taking the Same Orbit Height and the Same Number of Satellites

h [km]	Constellation	Optimal Constellation Parameters	η_{gc}	η_{isl}	η
16,519	Walker	$T = 6, P = 2, F = 0, i = 1.06$ rad	86%	95%	90%
	Flower	$N_p = 5, N_d = 2, N_s = 6, F_n = 18, F_d = 39, F_h = 0, i = 1.0$ rad	88%	98%	92%
13,899	Walker	$T = 8, P = 4, F = 1, i = 1.02$ rad	95%	86%	91%
	Flower	$N_p = 3, N_d = 1, N_s = 8, F_n = 3, F_d = 20, F_h = 0, i = 1.04$ rad	90%	97%	93%
10,361	Walker	$T = 10, P = 5, F = 1, i = 1.0$ rad	94%	81%	87%
	Flower	$N_p = 4, N_d = 1, N_s = 10, F_n = 28, F_d = 31, F_h = 0, i = 1.13$ rad	89%	89%	89%
8,049	Walker	$T = 12, P = 3, F = 2, i = 1.13$ rad	92%	77%	84%
	Flower	$N_p = 5, N_d = 1, N_s = 12, F_n = 8, F_d = 29, F_h = 0, i = 1.05$ rad	94%	76%	85%
6,398	Walker	$T = 16, P = 4, F = 1, i = 1.06$ rad	93%	69%	81%
	Flower	$N_p = 6, N_d = 1, N_s = 16, F_n = 24, F_d = 31, F_h = 0, i = 1.1$ rad	97%	69%	83%
5,151	Walker	$T = 22, P = 11, F = 5, i = 1.24$ rad	92%	62%	77%
	Flower	$N_p = 7, N_d = 1, N_s = 22, F_n = 3, F_d = 25, F_h = 0, i = 1.19$ rad	94%	62%	78%

which allows a more accurate phasing of satellites and a larger set of design solutions. Furthermore, a more uniform distribution of satellites around the Earth and a less time-varying relative position between satellites are provided with FCs. As an illustrative example, Fig. 4 shows the distribution of satellites for the optimized Walker Constellations and FCs at the orbit height $h = 8049$ km (as reported in Table 1). From this figure we can notice that the satellite distribution of the FC is more uniform and, hence, global performance in terms of coverage and ISLs availability is easily achieved. As a final consequence, in the optimization of FCs, there is a lower conflict between the two metrics η_{gc} and η_{isl} and, hence, a better tradeoff can be achieved.

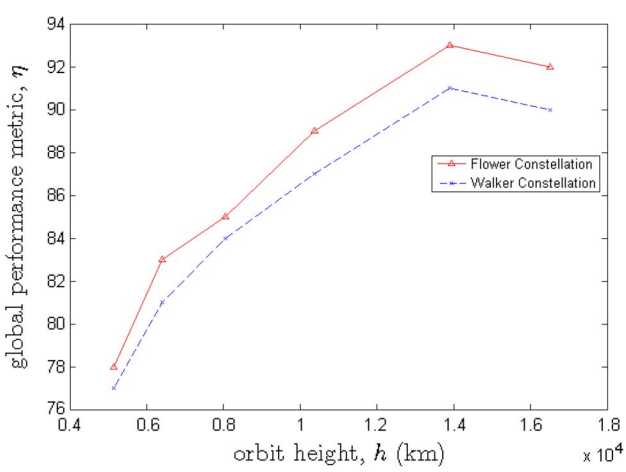

Fig. 3. ***Comparison of the trend of the performance metric η between Flower and Walker Constellations.***

IV. SYSTEM ENGINEERING: ORBIT PERTURBATIONS, DEPLOYMENT, AND MAINTENANCE

A. Orbit Perturbations

Whatever solution is adopted for the design of a satellite constellation, we need to manage the complexity of a system composed of many satellites with all the problems related to: launch of two or more satellites, distribution along the orbit with accurate positions, control and maintenance for all the lifespan, provision of the disposal at the end of life, and so on. Among all those items, one of the most challenging is the control and maintenance of the constellation because it deals with managing and monitoring a high number of satellites. However, apart from the technical aspect, the adopted control strategy is strictly related to the overall costs of the constellation. Therefore, it is particularly important to identify advantages and disadvantages of a given satellite constellation with respect to the control and maintenance in order to find out control elements of costs.

If satellites were subjected only to the gravitational field generated by a spherically symmetrical Earth, a constellation would follow the basic dynamics laws (two-body problem) and would keep its pattern for all the lifespan without the need of intervention from the ground station. Unfortunately, our planet, even if it has a spherical-like shape, presents different mismatches such as pole flattening and nonuniform mass distribution; actually, the Earth, if considered at enough

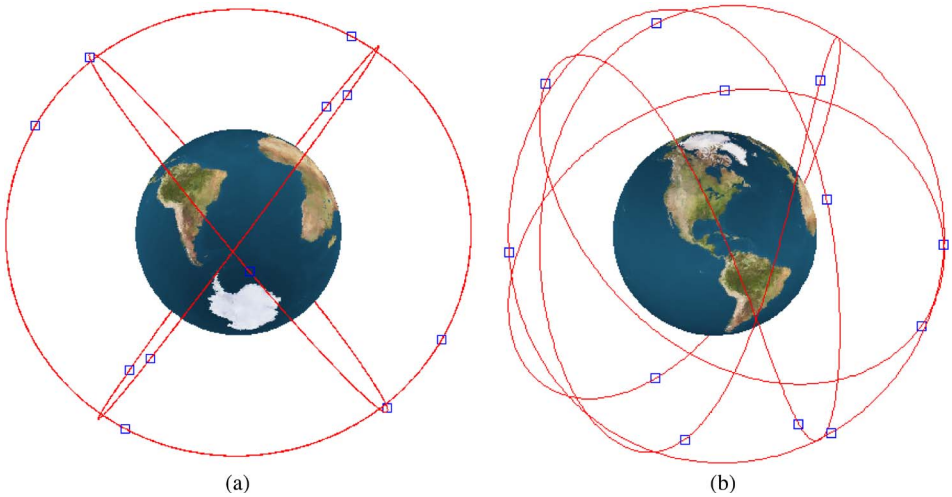

Fig. 4. ***Distribution of satellites for optimized Walker and Flower Constellations. (a) Distribution of satellites and orbits for an optimized Walker Constellation at height $h = 8049$ km (ECI reference frame). (b) Distribution of satellites and relative path for an optimized Flower Constellation at height $h = 8049$ km (ECEF reference frame).***

resolution, is similar to a smooth cocoon. In addition, the Earth is influenced by the gravitational field of other bodies, especially the Sun and the Moon; the first one because of the huge mass and the second one because of the relatively low distance. Other effects need to be considered, mainly for some low-altitude orbits, as the presence of atmosphere, which causes a slowing of the satellite with orbital decaying, and the solar radiation pressure, which modifies the orbital eccentricity, to be considered for higher orbits.

All these effects make the Kepler laws, exact solution of the two-body problem, accurate to describe the satellites motion just at a first approximation. Actually, just in case of a perfect spherically symmetrical Earth, Kepler laws allow to determine the orbit of a body (natural or artificial) by means of the six orbital elements. Mismatches from the ideal sphere and the presence of additive effects, previously mentioned, add perturbations to the orbital elements, causing time variations and therefore deviations from the Kepler solution. These perturbations appear as short/long period variations (with respect to the orbital period) and secular variations (which increase or decrease with time) in the orbital elements.

When the satellites are more than one, the perturbations effect must be seriously taken into account because it affects not just single satellites but also the structure and the shape of the initial constellation, causing a performance degradation of the service.

To keep the pattern of a constellation over a long time is a difficult task and the efficiency of accomplishing this objective is strictly related to the overall costs. Usually, control and maintenance of a constellation requires knowledge and control of each satellite but also control of each satellite with respect to the others.

B. Perturbations in Flower and Walker Constellations

Taking into account previous considerations, perturbations are of paramount importance in managing constellations.

In this respect, it is useful to compare FCs to the classical Walker Constellations (considering similar characteristics) in terms of perturbations evolution.

The Walker Constellation is characterized by an intrinsic symmetry due to the methodology of designing. On the other hand, the use of elliptical orbit FCs allows focused coverage over specific regions. However, elliptical orbits are more sensitive to perturbations.

For the comparison, we report as an example a case study related to Flower and Walker Constellations expressly studied for telemedicine applications [18]. Both have been built so that a suitable comparison was possible between them (as the same number of satellites, the same mean height, etc.). Telemedicine applications are a particular type of telecommunication service which can exploit real-time and/or store-and-forward applications, enabling the communication and sharing of medical information in electronic form, and thus facilitating access to remote expertise.

The metric used to compare the perturbation effects on the constellations has been the estimate of the Keplerian orbital elements drift on a time period of one year.

Usually, control and maintenance of satellites is expressed in terms of ΔV, which is the required velocity change to be applied to compensate for drifts in order to "fix" the satellite in the foreseen position. In general, a ΔV budget, sum of the velocity changes required during all the constellation lifespan, is calculated to take into account

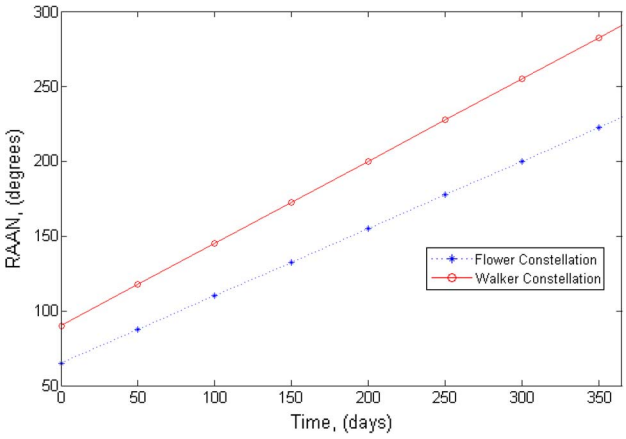

Fig. 5. ***RAAN evolution for one satellite of the Flower Constellation and the Walker Constellation.***

the fuel consumption due to orbit transfer, station keeping, orbit maneuvers, rephasing, plane change, disposal at end of life, and so on. This quantity, being directly related to fuel consumption and hence to the cost, allows to evaluate the influence of the control and maintenance over the overall cost.

Both constellations have been analyzed to evaluate the cumulative perturbations affecting the constellations over a one year period. In particular, gravitational geopotential (until 12th order and degree), Sun and Moon gravitational field and atmospheric drag (using the 1976 Standard Atmospheric Model) have been considered. For each constellation, the evolution of the six orbital elements has been analyzed.

Both constellations are affected mostly by the perturbations due to the J_2 effect, the value of which is three orders of magnitude higher than the following geopotential coefficients; it is also dominant with respect to the other effects because of the major proximity to the Earth. The J_2 effect produces secular variations in three orbital elements: the argument of perigee (ω), the right ascension of the ascending node (Ω), and the mean anomaly M.

The result of the perturbations is that the repeating ground track shifts in time due to the J_2 effect with respect to that one without perturbations. This behavior is mainly due to RAAN drift. Fig. 5 plots the RAAN drift of one satellite of the Flower and Walker Constellations over a one year period. The RAAN drift is similar to the drift of other satellites. As highlighted in this figure, the FC RAAN drift increases linearly and is about 165° for each satellite after one year of propagation. On the opposite, the Walker Constellation has a faster rise in RAAN with a maximum value of about 200° for each satellite.

The difference is significant and affects the most perturbed element in both constellations. This means a significant difference in the ΔV budget required to control the constellation configuration. This control must be operated on four orbital planes for the Walker Constellations and eight for the FCs (for this case study); in both cases, the control must be applied on eight satellites.

C. Deployment Strategy Considerations

One of the issues to deal with when satellite constellations are considered is the deployment. The launch and deployment on the operating orbits of two or more satellites is a difficult and especially costly task, and therefore, it is important to optimize the adopted strategy. The final configuration of a constellation (usually achieved through multiple launches) is almost always composed by different orbital planes, so that the effective deployment can follow two main options:

- direct injection of each satellite in its assigned orbital slot;
- injection of the global cluster in a transfer orbital plane not necessarily belonging to the final configuration and following a transfer to the assigned slot by means of single spacecraft propulsion capabilities.

The first option is certainly possible with modern launchers, often providing a number (usually $5 < n < 10$) of re-ignitions, and therefore, able to achieve $n + 1$ different dedicated deliveries. However, the second opportunity will be shortly investigated in the following.

Note that all the considerations below should then be matched with the performance offered by the launcher at the different altitudes, in order to identify the optimal solution. The acquisition includes an in-plane maneuver (slot acquisition), really similar to the rephasing maneuver, which is carried out from time to time to counteract the decay originated by the atmospheric drag, and an out-of-plane maneuver. This second maneuver is by far the most expensive as far as the energetic contribution (ΔV) is concerned and will be therefore considered in detail as it is the driver for the global injection phase. Assuming that the injection could be carried out at the critical inclination (63.4°), the only out-of-plane parameter to correct is the RAAN (Ω). The RAAN can be modified:

- by means of a direct maneuver, carried on at the locations with a difference of ±90° with respect to the node, with the following relation holding for the tradeoff between the effect and the cost:

$$\Delta\Omega = \frac{1}{\sin i}\frac{\Delta V}{v_\theta} \tag{13}$$

  where v_θ is the tangential component of the orbital velocity; from that relation it implies that the more convenient location for such a maneuver is as close as possible to the apogee (the apogee itself will be a possible choice only if $\omega = 90°$);
- via a transfer of the spacecraft on intermediate orbits (drift orbits) between the injection and the

final ones; these orbits will have a drift, originated by the oblateness of the Earth, and specifically by the dominating term J_2, which is equal to (for each revolution)

$$\Delta\Omega_{\text{orb}} = -3\pi J_2 \frac{R_{\text{mean}}^2}{a^2(1-e^2)} \cos i. \tag{14}$$

Therefore, there is the possibility to obtain the desired spacing of the planes by acting on the semimajor axis or eccentricity via a double in-plane maneuver: the first one moving to the intermediate, and the second one returning to the correct semimajor axis/inclination values once the separation has been achieved. Of course, the satellite targeted to the same orbital plane of injection will not face this maneuver.

D. Control and Maintenance Solutions

Elliptic FCs which do not use the critical inclination, i.e., $i = 63.4°$ or $i = 116.6°$ must consider the perigee drift. Concerning constellation control and maintenance, the maintenance of the perigee is an important item to be considered. Such a drift is problematic for applications as telecommunications, for example, requiring the apogee fixed over a specific region of the Earth. Equation (15) expresses the perigee drift in terms of radial and transverse acceleration a_r and a_s, respectively

$$\frac{d\omega}{dt} = -\frac{\sqrt{1-e^2}\cos\phi}{nae} a_r + \frac{\pi \sin\phi}{eh}\left(\frac{2+e\cos\phi}{1+e\cos\phi}\right) a_s \tag{15}$$

where n, ϕ, p are, respectively, mean motion, true anomaly, and $p = a(1-e^2)$.

In order to compensate the perigee drift there are two options: tangential thrust at $\phi = 90°, 270°$ or radial thrust at $\phi = 0°, 180°$. In case of tangential thrust, choosing $\phi = 90°, 270°$ from (15), we get

$$\frac{d\omega}{dt} = -\frac{p}{eh}\sin\phi \frac{2+e\cos\phi}{1+e\cos\phi} a_s \tag{16}$$

and afterwards

$$\Delta\Omega_{\text{orbit}} = \frac{2\pi}{eh}\Delta V_{\text{orbit}}. \tag{17}$$

Putting this expression equal to the orbit drift

$$\Delta\Omega_{\text{orbit}} = -\frac{3\pi J_2}{2}\frac{R^2}{p^2}(5\cos^2 i - 1) \tag{18}$$

we get the ΔV required to compensate for the drift

$$\Delta V_{\text{orbit}} = -e\sqrt{\frac{\mu}{p}}\frac{3\pi J_2}{4}\frac{R^2}{p^2}(5\cos^2 i - 1). \tag{19}$$

In this case, the tangential impulse allows to control the perigee with a certain ΔV but at the same time changes semimajor axis and eccentricity. In case of radial thrust, from (15), we get

$$\Delta\omega_{\text{orbit}} = \frac{1}{e}\sqrt{\frac{p}{\mu}}\Delta V_{\text{orbit}} \tag{20}$$

then, considering the orbit drift, we get

$$\Delta V_{\text{orbit}} = -e\sqrt{\frac{\mu}{p}}\frac{3\pi J_2}{2}\frac{R^2}{p^2}(5\cos^2 i - 1). \tag{21}$$

In this case, the ΔV required for the perigee maintenance is twice the one used for the tangential impulse but it has the advantage of not affecting semimajor axis and eccentricity.

V. CONCLUSION

The design of a satellite constellation for telecommunication services is a very complex task that encompasses: the definition of system requirements/metrics/costs, the choice of the design method (Walker, Flower, etc.) and the optimization tool.

The constellation design method based on the relatively new FC theory provides new mathematical tools which increases the solution space.

In this paper, after a brief story of the FC theory, we used this theory to optimize the design of the constellation and to compare the classical design method represented by the Walker scheme with the Flower scheme. We have shown that the FC design method provides enhanced performance due to the intrinsic higher number of possible configurations which allows a more accurate phasing of the satellites.

Furthermore, the comparison between Flower and Walker Constellations in terms of system costs for orbit control and deployment justifies the final selection of FCs. ■

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Mauro De Sanctis received the "Laurea" degree in telecommunications engineering and the Ph.D. degree in telecommunications and microelectronics engineering from the University of Roma "Tor Vergata," Rome, Italy, in 2002 and 2005, respectively.

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